

```

/* USER CODE BEGIN Header */
/**
*****

* @file      : main.c
* @brief     : Main program body
*****

* @attention

*

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*

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*
*****

*/

/* USER CODE END Header */

/* Includes -----*/

#include "main.h"


/* Private includes -----*/

/* USER CODE BEGIN Includes */

#include <stdio.h>
#include "stm32f4xx.h"
#include "lcd_stm32f4.h"

/* USER CODE END Includes */


/* Private typedef -----*/

/* USER CODE BEGIN PTD */


/* USER CODE END PTD */


/* Private define -----*/

/* USER CODE BEGIN PD */

// TODO: Add values for below variables

#define NS          1024          // Number of samples in LUT

#define NS_WAVE     8192          // Number of samples in LUT

```

```

#define TIM2CLK    16000000    // STM Clock frequency: Hint You might want to check the ioc file
#define F_SIGNAL    1          // Frequency of output analog signal

/* USER CODE END PD */

/* Private macro -----*/
/* USER CODE BEGIN PM */

/* USER CODE END PM */

/* Private variables -----*/
TIM_HandleTypeDef htim2;
TIM_HandleTypeDef htim3;
DMA_HandleTypeDef hdma_tim2_ch1;
uint32_t current_tick;
uint32_t last_button_press;
uint8_t current_waveform;
uint32_t *selected_lut;
char *waveform_name;
double frequency;
uint32_t ns_to_use;

/* USER CODE BEGIN PV */
// TODO: Add code for global variables, including LUTs
// Sin Wave LUT: 1024 samples, 4096 bytes (4.00KB)
uint32_t Sin_LUT[NS] = {
    2048, 2061, 2073, 2086, 2098, 2111, 2123, 2136, 2148, 2161, 2174, 2186, 2199, 2211, 2224,
    2236,
    2249, 2261, 2274, 2286, 2299, 2311, 2323, 2336, 2348, 2361, 2373, 2386, 2398, 2410, 2423,
    2435,
    2447, 2460, 2472, 2484, 2497, 2509, 2521, 2533, 2545, 2558, 2570, 2582, 2594, 2606, 2618,
    2630,
    2642, 2654, 2666, 2678, 2690, 2702, 2714, 2726, 2738, 2749, 2761, 2773, 2785, 2796, 2808,
    2820,
    2831, 2843, 2855, 2866, 2878, 2889, 2900, 2912, 2923, 2935, 2946, 2957, 2968, 2980, 2991,
    3002,
    3013, 3024, 3035, 3046, 3057, 3068, 3079, 3090, 3100, 3111, 3122, 3133, 3143, 3154, 3164,
    3175,
    3185, 3196, 3206, 3216, 3227, 3237, 3247, 3257, 3267, 3277, 3287, 3297, 3307, 3317, 3327,
    3337,

```

3347, 3356, 3366, 3376, 3385, 3395, 3404, 3413, 3423, 3432, 3441, 3450, 3459, 3469, 3478, 3487,

3495, 3504, 3513, 3522, 3531, 3539, 3548, 3556, 3565, 3573, 3581, 3590, 3598, 3606, 3614, 3622,

3630, 3638, 3646, 3654, 3662, 3669, 3677, 3685, 3692, 3700, 3707, 3714, 3722, 3729, 3736, 3743,

3750, 3757, 3764, 3771, 3777, 3784, 3791, 3797, 3804, 3810, 3817, 3823, 3829, 3835, 3841, 3847,

3853, 3859, 3865, 3871, 3876, 3882, 3888, 3893, 3898, 3904, 3909, 3914, 3919, 3924, 3929, 3934,

3939, 3944, 3949, 3953, 3958, 3962, 3967, 3971, 3975, 3980, 3984, 3988, 3992, 3996, 3999, 4003,

4007, 4010, 4014, 4017, 4021, 4024, 4027, 4031, 4034, 4037, 4040, 4042, 4045, 4048, 4051, 4053,

4056, 4058, 4060, 4063, 4065, 4067, 4069, 4071, 4073, 4075, 4076, 4078, 4080, 4081, 4083, 4084,

4085, 4086, 4087, 4088, 4089, 4090, 4091, 4092, 4093, 4093, 4094, 4094, 4094, 4095, 4095, 4095,

4095, 4095, 4095, 4095, 4094, 4094, 4094, 4093, 4093, 4092, 4091, 4090, 4089, 4088, 4087, 4086,

4085, 4084, 4083, 4081, 4080, 4078, 4076, 4075, 4073, 4071, 4069, 4067, 4065, 4063, 4060, 4058,

4056, 4053, 4051, 4048, 4045, 4042, 4040, 4037, 4034, 4031, 4027, 4024, 4021, 4017, 4014, 4010,

4007, 4003, 3999, 3996, 3992, 3988, 3984, 3980, 3975, 3971, 3967, 3962, 3958, 3953, 3949, 3944,

3939, 3934, 3929, 3924, 3919, 3914, 3909, 3904, 3898, 3893, 3888, 3882, 3876, 3871, 3865, 3859,

3853, 3847, 3841, 3835, 3829, 3823, 3817, 3810, 3804, 3797, 3791, 3784, 3777, 3771, 3764, 3757,

3750, 3743, 3736, 3729, 3722, 3714, 3707, 3700, 3692, 3685, 3677, 3669, 3662, 3654, 3646, 3638,

3630, 3622, 3614, 3606, 3598, 3590, 3581, 3573, 3565, 3556, 3548, 3539, 3531, 3522, 3513, 3504,

3495, 3487, 3478, 3469, 3459, 3450, 3441, 3432, 3423, 3413, 3404, 3395, 3385, 3376, 3366, 3356,

3347, 3337, 3327, 3317, 3307, 3297, 3287, 3277, 3267, 3257, 3247, 3237, 3227, 3216, 3206, 3196,

3185, 3175, 3164, 3154, 3143, 3133, 3122, 3111, 3100, 3090, 3079, 3068, 3057, 3046, 3035, 3024,

3013, 3002, 2991, 2980, 2968, 2957, 2946, 2935, 2923, 2912, 2900, 2889, 2878, 2866, 2855, 2843,

2831, 2820, 2808, 2796, 2785, 2773, 2761, 2749, 2738, 2726, 2714, 2702, 2690, 2678, 2666, 2654,

2642, 2630, 2618, 2606, 2594, 2582, 2570, 2558, 2545, 2533, 2521, 2509, 2497, 2484, 2472, 2460,

2447, 2435, 2423, 2410, 2398, 2386, 2373, 2361, 2348, 2336, 2323, 2311, 2299, 2286, 2274, 2261,

2249, 2236, 2224, 2211, 2199, 2186, 2174, 2161, 2148, 2136, 2123, 2111, 2098, 2086, 2073,
2061,
2048, 2035, 2023, 2010, 1998, 1985, 1973, 1960, 1948, 1935, 1922, 1910, 1897, 1885, 1872,
1860,
1847, 1835, 1822, 1810, 1797, 1785, 1773, 1760, 1748, 1735, 1723, 1710, 1698, 1686, 1673,
1661,
1649, 1636, 1624, 1612, 1599, 1587, 1575, 1563, 1551, 1538, 1526, 1514, 1502, 1490, 1478,
1466,
1454, 1442, 1430, 1418, 1406, 1394, 1382, 1370, 1358, 1347, 1335, 1323, 1311, 1300, 1288,
1276,
1265, 1253, 1241, 1230, 1218, 1207, 1196, 1184, 1173, 1161, 1150, 1139, 1128, 1116, 1105,
1094,
1083, 1072, 1061, 1050, 1039, 1028, 1017, 1006, 996, 985, 974, 963, 953, 942, 932,
921,
911, 900, 890, 880, 869, 859, 849, 839, 829, 819, 809, 799, 789, 779, 769,
759,
749, 740, 730, 720, 711, 701, 692, 683, 673, 664, 655, 646, 637, 627, 618,
609,
601, 592, 583, 574, 565, 557, 548, 540, 531, 523, 515, 506, 498, 490, 482,
474,
466, 458, 450, 442, 434, 427, 419, 411, 404, 396, 389, 382, 374, 367, 360,
353,
346, 339, 332, 325, 319, 312, 305, 299, 292, 286, 279, 273, 267, 261, 255,
249,
243, 237, 231, 225, 220, 214, 208, 203, 198, 192, 187, 182, 177, 172, 167,
162,
157, 152, 147, 143, 138, 134, 129, 125, 121, 116, 112, 108, 104, 100, 97,
93,
89, 86, 82, 79, 75, 72, 69, 65, 62, 59, 56, 54, 51, 48, 45,
43,
40, 38, 36, 33, 31, 29, 27, 25, 23, 21, 20, 18, 16, 15, 13,
12,
11, 10, 9, 8, 7, 6, 5, 4, 3, 3, 2, 2, 2, 1, 1,
1,
1, 1, 1, 1, 2, 2, 2, 3, 3, 4, 5, 6, 7, 8, 9,
10,
11, 12, 13, 15, 16, 18, 20, 21, 23, 25, 27, 29, 31, 33, 36,
38,
40, 43, 45, 48, 51, 54, 56, 59, 62, 65, 69, 72, 75, 79, 82,
86,
89, 93, 97, 100, 104, 108, 112, 116, 121, 125, 129, 134, 138, 143, 147,
152,
157, 162, 167, 172, 177, 182, 187, 192, 198, 203, 208, 214, 220, 225, 231,
237,
243, 249, 255, 261, 267, 273, 279, 286, 292, 299, 305, 312, 319, 325, 332,
339,
346, 353, 360, 367, 374, 382, 389, 396, 404, 411, 419, 427, 434, 442, 450,
458,

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        466, 474, 482, 490, 498, 506, 515, 523, 531, 540, 548, 557, 565, 574, 583,
592,

        601, 609, 618, 627, 637, 646, 655, 664, 673, 683, 692, 701, 711, 720, 730,
740,

        749, 759, 769, 779, 789, 799, 809, 819, 829, 839, 849, 859, 869, 880, 890,
900,

        911, 921, 932, 942, 953, 963, 974, 985, 996, 1006, 1017, 1028, 1039, 1050, 1061,
1072,

        1083, 1094, 1105, 1116, 1128, 1139, 1150, 1161, 1173, 1184, 1196, 1207, 1218, 1230, 1241,
1253,

        1265, 1276, 1288, 1300, 1311, 1323, 1335, 1347, 1358, 1370, 1382, 1394, 1406, 1418, 1430,
1442,

        1454, 1466, 1478, 1490, 1502, 1514, 1526, 1538, 1551, 1563, 1575, 1587, 1599, 1612, 1624,
1636,

        1649, 1661, 1673, 1686, 1698, 1710, 1723, 1735, 1748, 1760, 1773, 1785, 1797, 1810, 1822,
1835,

        1847, 1860, 1872, 1885, 1897, 1910, 1922, 1935, 1948, 1960, 1973, 1985, 1998, 2010, 2023,
2035

};

// Saw Wave LUT: 1024 samples, 4096 bytes (4.00KB)

uint32_t Saw_LUT[NS] = {

        0,    4,    8,   12,   16,   20,   24,   28,   32,   36,   40,   44,   48,   52,   56,
60,

        64,   68,   72,   76,   80,   84,   88,   92,   96,  100,  104,  108,  112,  116,  120,
124,

        128,  132,  136,  140,  144,  148,  152,  156,  160,  164,  168,  172,  176,  180,  184,
188,

        192,  196,  200,  204,  208,  212,  216,  220,  224,  228,  232,  236,  240,  244,  248,
252,

        256,  260,  264,  268,  272,  276,  280,  284,  288,  292,  296,  300,  304,  308,  312,
316,

        320,  324,  328,  332,  336,  340,  344,  348,  352,  356,  360,  364,  368,  372,  376,
380,

        384,  388,  392,  396,  400,  404,  408,  412,  416,  420,  424,  428,  432,  436,  440,
444,

        448,  452,  456,  460,  464,  468,  472,  476,  480,  484,  488,  492,  496,  500,  504,
508,

        512,  516,  520,  524,  528,  532,  536,  540,  544,  548,  552,  556,  560,  564,  568,
572,

        576,  580,  584,  588,  592,  596,  600,  604,  608,  612,  616,  620,  624,  628,  632,
636,

        640,  644,  648,  652,  656,  660,  664,  668,  672,  676,  680,  684,  688,  692,  696,
700,

        704,  708,  712,  716,  720,  724,  728,  732,  736,  740,  744,  748,  752,  756,  760,
764,

        768,  772,  776,  780,  784,  788,  792,  796,  800,  804,  808,  812,  816,  820,  824,
828,

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832, 836, 840, 844, 848, 852, 856, 860, 864, 868, 872, 876, 880, 884, 888,
892,
896, 900, 904, 908, 912, 916, 920, 924, 928, 932, 936, 940, 944, 948, 952,
956,
960, 964, 968, 972, 976, 980, 984, 988, 992, 996, 1000, 1004, 1008, 1012, 1016,
1020,
1024, 1028, 1032, 1036, 1040, 1044, 1048, 1052, 1056, 1060, 1064, 1068, 1072, 1076, 1080,
1084,
1088, 1092, 1096, 1100, 1104, 1108, 1112, 1116, 1120, 1124, 1128, 1132, 1136, 1140, 1144,
1148,
1152, 1156, 1160, 1164, 1168, 1172, 1176, 1180, 1184, 1188, 1192, 1196, 1200, 1204, 1208,
1212,
1216, 1220, 1224, 1228, 1232, 1236, 1240, 1244, 1248, 1252, 1256, 1260, 1264, 1268, 1272,
1276,
1280, 1284, 1288, 1292, 1296, 1300, 1304, 1308, 1312, 1316, 1320, 1324, 1328, 1332, 1336,
1340,
1344, 1348, 1352, 1356, 1360, 1364, 1368, 1372, 1376, 1380, 1384, 1388, 1392, 1396, 1400,
1404,
1408, 1412, 1416, 1420, 1424, 1428, 1432, 1436, 1440, 1444, 1448, 1452, 1456, 1460, 1464,
1468,
1472, 1476, 1480, 1484, 1488, 1492, 1496, 1500, 1504, 1508, 1512, 1516, 1520, 1524, 1528,
1532,
1536, 1540, 1544, 1548, 1552, 1556, 1560, 1564, 1568, 1572, 1576, 1580, 1584, 1588, 1592,
1596,
1600, 1604, 1608, 1612, 1616, 1620, 1624, 1628, 1632, 1636, 1640, 1644, 1648, 1652, 1656,
1660,
1664, 1668, 1672, 1676, 1680, 1684, 1688, 1692, 1696, 1700, 1704, 1708, 1712, 1716, 1720,
1724,
1728, 1732, 1736, 1740, 1744, 1748, 1752, 1756, 1760, 1764, 1768, 1772, 1776, 1780, 1784,
1788,
1792, 1796, 1800, 1804, 1808, 1812, 1816, 1820, 1824, 1828, 1832, 1836, 1840, 1844, 1848,
1852,
1856, 1860, 1864, 1868, 1872, 1876, 1880, 1884, 1888, 1892, 1896, 1900, 1904, 1908, 1912,
1916,
1920, 1924, 1928, 1932, 1936, 1940, 1944, 1948, 1952, 1956, 1960, 1964, 1968, 1972, 1976,
1980,
1984, 1988, 1992, 1996, 2000, 2004, 2008, 2012, 2016, 2020, 2024, 2028, 2032, 2036, 2040,
2044,
2048, 2052, 2056, 2060, 2064, 2068, 2072, 2076, 2080, 2084, 2088, 2092, 2096, 2100, 2104,
2108,
2112, 2116, 2120, 2124, 2128, 2132, 2136, 2140, 2144, 2148, 2152, 2156, 2160, 2164, 2168,
2172,
2176, 2180, 2184, 2188, 2192, 2196, 2200, 2204, 2208, 2212, 2216, 2220, 2224, 2228, 2232,
2236,
2240, 2244, 2248, 2252, 2256, 2260, 2264, 2268, 2272, 2276, 2280, 2284, 2288, 2292, 2296,
2300,
2304, 2308, 2312, 2316, 2320, 2324, 2328, 2332, 2336, 2340, 2344, 2348, 2352, 2356, 2360,
2364,

2368, 2372, 2376, 2380, 2384, 2388, 2392, 2396, 2400, 2404, 2408, 2412, 2416, 2420, 2424, 2428,

2432, 2436, 2440, 2444, 2448, 2452, 2456, 2460, 2464, 2468, 2472, 2476, 2480, 2484, 2488, 2492,

2496, 2500, 2504, 2508, 2512, 2516, 2520, 2524, 2528, 2532, 2536, 2540, 2544, 2548, 2552, 2556,

2560, 2564, 2568, 2572, 2576, 2580, 2584, 2588, 2592, 2596, 2600, 2604, 2608, 2612, 2616, 2620,

2624, 2628, 2632, 2636, 2640, 2644, 2648, 2652, 2656, 2660, 2664, 2668, 2672, 2676, 2680, 2684,

2688, 2692, 2696, 2700, 2704, 2708, 2712, 2716, 2720, 2724, 2728, 2732, 2736, 2740, 2744, 2748,

2752, 2756, 2760, 2764, 2768, 2772, 2776, 2780, 2784, 2788, 2792, 2796, 2800, 2804, 2808, 2812,

2816, 2820, 2824, 2828, 2832, 2836, 2840, 2844, 2848, 2852, 2856, 2860, 2864, 2868, 2872, 2876,

2880, 2884, 2888, 2892, 2896, 2900, 2904, 2908, 2912, 2916, 2920, 2924, 2928, 2932, 2936, 2940,

2944, 2948, 2952, 2956, 2960, 2964, 2968, 2972, 2976, 2980, 2984, 2988, 2992, 2996, 3000, 3004,

3008, 3012, 3016, 3020, 3024, 3028, 3032, 3036, 3040, 3044, 3048, 3052, 3056, 3060, 3064, 3068,

3072, 3076, 3080, 3084, 3088, 3092, 3096, 3100, 3104, 3108, 3112, 3116, 3120, 3124, 3128, 3132,

3136, 3140, 3144, 3148, 3152, 3156, 3160, 3164, 3168, 3172, 3176, 3180, 3184, 3188, 3192, 3196,

3200, 3204, 3208, 3212, 3216, 3220, 3224, 3228, 3232, 3236, 3240, 3244, 3248, 3252, 3256, 3260,

3264, 3268, 3272, 3276, 3280, 3284, 3288, 3292, 3296, 3300, 3304, 3308, 3312, 3316, 3320, 3324,

3328, 3332, 3336, 3340, 3344, 3348, 3352, 3356, 3360, 3364, 3368, 3372, 3376, 3380, 3384, 3388,

3392, 3396, 3400, 3404, 3408, 3412, 3416, 3420, 3424, 3428, 3432, 3436, 3440, 3444, 3448, 3452,

3456, 3460, 3464, 3468, 3472, 3476, 3480, 3484, 3488, 3492, 3496, 3500, 3504, 3508, 3512, 3516,

3520, 3524, 3528, 3532, 3536, 3540, 3544, 3548, 3552, 3556, 3560, 3564, 3568, 3572, 3576, 3580,

3584, 3588, 3592, 3596, 3600, 3604, 3608, 3612, 3616, 3620, 3624, 3628, 3632, 3636, 3640, 3644,

3648, 3652, 3656, 3660, 3664, 3668, 3672, 3676, 3680, 3684, 3688, 3692, 3696, 3700, 3704, 3708,

3712, 3716, 3720, 3724, 3728, 3732, 3736, 3740, 3744, 3748, 3752, 3756, 3760, 3764, 3768, 3772,

3776, 3780, 3784, 3788, 3792, 3796, 3800, 3804, 3808, 3812, 3816, 3820, 3824, 3828, 3832, 3836,

3840, 3844, 3848, 3852, 3856, 3860, 3864, 3868, 3872, 3876, 3880, 3884, 3888, 3892, 3896, 3900,

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    3904, 3908, 3912, 3916, 3920, 3924, 3928, 3932, 3936, 3940, 3944, 3948, 3952, 3956, 3960,
    3964,

    3968, 3972, 3976, 3980, 3984, 3988, 3992, 3996, 4000, 4004, 4008, 4012, 4016, 4020, 4024,
    4028,

    4032, 4036, 4040, 4044, 4048, 4052, 4056, 4060, 4064, 4068, 4072, 4076, 4080, 4084, 4088,
    4092

};

// Triangle Wave LUT: 1024 samples, 4096 bytes (4.00KB)
uint32_t Triangle_LUT[NS] = {
    0,    8,   16,   24,   32,   40,   48,   56,   64,   72,   80,   88,   96,  104,  112,
120,

    128,  136,  144,  152,  160,  168,  176,  184,  192,  200,  208,  216,  224,  232,  240,
248,

    256,  264,  272,  280,  288,  296,  304,  312,  320,  328,  336,  344,  352,  360,  368,
376,

    384,  392,  400,  408,  416,  424,  432,  440,  448,  456,  464,  472,  480,  488,  496,
504,

    512,  520,  528,  536,  544,  552,  560,  568,  576,  584,  592,  600,  608,  616,  624,
632,

    640,  648,  656,  664,  672,  680,  688,  696,  704,  712,  720,  728,  736,  744,  752,
760,

    768,  776,  784,  792,  800,  808,  816,  824,  832,  840,  848,  856,  864,  872,  880,
888,

    896,  904,  912,  920,  928,  936,  944,  952,  960,  968,  976,  984,  992, 1000, 1008,
1016,

    1024, 1032, 1040, 1048, 1056, 1064, 1072, 1080, 1088, 1096, 1104, 1112, 1120, 1128, 1136,
1144,

    1152, 1160, 1168, 1176, 1184, 1192, 1200, 1208, 1216, 1224, 1232, 1240, 1248, 1256, 1264,
1272,

    1280, 1288, 1296, 1304, 1312, 1320, 1328, 1336, 1344, 1352, 1360, 1368, 1376, 1384, 1392,
1400,

    1408, 1416, 1424, 1432, 1440, 1448, 1456, 1464, 1472, 1480, 1488, 1496, 1504, 1512, 1520,
1528,

    1536, 1544, 1552, 1560, 1568, 1576, 1584, 1592, 1600, 1608, 1616, 1624, 1632, 1640, 1648,
1656,

    1664, 1672, 1680, 1688, 1696, 1704, 1712, 1720, 1728, 1736, 1744, 1752, 1760, 1768, 1776,
1784,

    1792, 1800, 1808, 1816, 1824, 1832, 1840, 1848, 1856, 1864, 1872, 1880, 1888, 1896, 1904,
1912,

    1920, 1928, 1936, 1944, 1952, 1960, 1968, 1976, 1984, 1992, 2000, 2008, 2016, 2024, 2032,
2040,

    2048, 2056, 2064, 2072, 2080, 2088, 2096, 2104, 2112, 2120, 2128, 2136, 2144, 2152, 2160,
2168,

    2176, 2184, 2192, 2200, 2208, 2216, 2224, 2232, 2240, 2248, 2256, 2264, 2272, 2280, 2288,
2296,

    2304, 2312, 2320, 2328, 2336, 2344, 2352, 2360, 2368, 2376, 2384, 2392, 2400, 2408, 2416,
2424,

```


2432, 2440, 2448, 2456, 2464, 2472, 2480, 2488, 2496, 2504, 2512, 2520, 2528, 2536, 2544, 2552,

2560, 2568, 2576, 2584, 2592, 2600, 2608, 2616, 2624, 2632, 2640, 2648, 2656, 2664, 2672, 2680,

2688, 2696, 2704, 2712, 2720, 2728, 2736, 2744, 2752, 2760, 2768, 2776, 2784, 2792, 2800, 2808,

2816, 2824, 2832, 2840, 2848, 2856, 2864, 2872, 2880, 2888, 2896, 2904, 2912, 2920, 2928, 2936,

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3934, 3922, 3953, 3935, 3695, 3335, 3153, 3198, 3364, 3246, 2740, 2292, 2086, 1904, 1927, 2120,

2204, 1975, 1629, 1646, 1942, 1830, 1780, 1363, 1118, 977, 1573, 2522, 2130, 2029, 1987, 1079,

800, 950, 966, 1434, 1019, 376, 811, 1217, 656, 1428, 846, 1169, 658, 655, 1208, 2015,

846, 1582, 1875, 1384, 1850, 1542, 2213, 1615, 1209, 2496, 2001, 1461, 2165, 2931, 2111, 2064,

2591, 3188, 3080, 2244, 2100, 1567, 2447, 2865, 2918, 3311, 3897, 3204, 3171, 3125, 3029, 3074,

3157, 3141, 3329, 2929, 2589, 2725, 3337, 2451, 1986, 2085, 1828, 2452, 1791, 1340, 1572, 2292,

1981, 2138, 1536, 2728, 2787, 2398, 1916, 1934, 1307, 931, 671, 794, 991, 880, 817, 304,

268, 590, 473, 444, 509, 743, 936, 944, 750, 1038, 1508, 2009, 2138, 2484, 2576, 2848,

2993, 2817, 3035, 2503, 2782, 2747, 2665, 2076, 2573, 2008, 2460, 2639, 2310, 2663, 2133, 2013,

1926, 1545, 2018, 1251, 1634, 1320, 1285, 1599, 1590, 1636, 1104, 1426, 1266, 1515, 1846, 1698,

1176, 2241, 2748, 2633, 2604, 2284, 2557, 1586, 1672, 1401, 2772, 2021, 2542, 1997, 1671, 1497,

1224, 2720, 2743, 2225, 1773, 923, 537, 1250, 2273, 1733, 1105, 1465, 1521, 1952, 2285, 1779,

1964, 2166, 2972, 1868, 3088, 2540, 1965, 2020, 2556, 2153, 1995, 1840, 1397, 1370, 1577, 1321,

1986, 1619, 1433, 1755, 1654, 1314, 1038, 1188, 1904, 2011, 2519, 2150, 2789, 2079, 2870, 2555,

2504, 2323, 2491, 2515, 2825, 2598, 2562, 2590, 2284, 2369, 1446, 2073, 1741, 1416, 1545, 1453,

2141, 1699, 2348, 2756, 2476, 2341, 2378, 2202, 2518, 2263, 1988, 1578, 1861, 1727, 2079, 2255,

2767, 2610, 2510, 2398, 2190, 2587, 2099, 2047, 1660, 1505, 1539, 1550, 2190, 1767, 1601, 2781,

```

    2563, 1345, 1543, 2428, 2881, 2936, 2502, 2205, 2256, 2474, 1922, 1843, 1908, 1490, 1481,
    1676,

    1208, 1424, 1336, 2084, 1896, 2170, 2341, 2396, 2615, 2681, 2334, 2041, 1990, 1873, 1343,
    1428,

    1570, 1534, 1787, 1567, 1810, 1992, 2263, 2513, 2095, 2091, 1618, 1564, 1543, 1703, 1646,
    1656,

    2030, 2253, 2420, 2239, 1944, 1798, 1630, 1738, 1884, 2273, 2559, 2321, 2610, 1945, 1866,
    1744,

    1686, 1964, 2169, 2519, 2440, 2399, 1956, 1724, 2222, 2465, 2295, 1882, 2181, 2274, 2027,
    2201

};

```

```

// TODO: Equation to calculate TIM2_Ticks

uint32_t TIM2_Ticks = TIM2CLK / (NS * F_SIGNAL); // How often to write new LUT value

uint32_t DestAddress = (uint32_t) &(TIM3->CCR3); // Write LUT TO TIM3->CCR3 to modify PWM duty
cycle

```

```

/* USER CODE END PV */

```

```

/* Private function prototypes -----*/

```

```

void SystemClock_Config(void);

static void MX_GPIO_Init(void);

static void MX_DMA_Init(void);

static void MX_TIM2_Init(void);

static void MX_TIM3_Init(void);

```

```

/* USER CODE BEGIN PFP */

```

```

void EXTI0_IRQHandler(void);

```

```

/* USER CODE END PFP */

```

```

/* Private user code -----*/

```

```

/* USER CODE BEGIN 0 */

```

```

/* USER CODE END 0 */

```

```

/**

```

```

 * @brief The application entry point.

```

```

 * @retval int

```

```

    */

int main(void)
{

    /* USER CODE BEGIN 1 */

    /* USER CODE END 1 */

    /* MCU Configuration-----*/

    /* Reset of all peripherals, Initializes the Flash interface and the Systick. */
    HAL_Init();

    /* USER CODE BEGIN Init */
    //  init_LCD();
    //  HAL_Delay(50); // Give LCD time to initialize
    /* USER CODE END Init */

    /* Configure the system clock */
    SystemClock_Config();

    /* USER CODE BEGIN SysInit */

    HAL_Delay(100);
    init_LCD();
    HAL_Delay(100);
    /* USER CODE END SysInit */

    /* Initialize all configured peripherals */
    MX_GPIO_Init();
    HAL_GPIO_WritePin(GPIOB, GPIO_PIN_4, GPIO_PIN_RESET);
    MX_DMA_Init();
    MX_TIM2_Init();
    MX_TIM3_Init();

    /* USER CODE BEGIN 2 */
    //  TODO: Start TIM3 in PWM mode on channel 3
    HAL_TIM_PWM_Start(&htim3, TIM_CHANNEL_3);

    //  TODO: Start TIM2 in Output Compare (OC) mode on channel 1
    HAL_TIM_OC_Start(&htim2, TIM_CHANNEL_1);

```

```
// TODO: Start DMA in IT mode on TIM2->CH1. Source is LUT and Dest is TIM3->CCR3; start with Sine LUT
```

```
HAL_DMA_Start_IT(&hdma_tim2_ch1, (uint32_t)Sin_LUT, DestAddress, NS);
```

```
// TODO: Write current waveform to LCD(Sine is the first waveform)
```

```
lcd_command(CLEAR);
```

```
lcd_putstr("Sine");
```

```
// TODO: Enable DMA (start transfer from LUT to CCR)
```

```
__HAL_TIM_ENABLE_DMA(&htim2, TIM_DMA_CC1);
```

```
/* USER CODE END 2 */
```

```
/* Infinite loop */
```

```
/* USER CODE BEGIN WHILE */
```

```
while (1)
```

```
{
```

```
    /* USER CODE END WHILE */
```

```
    /* USER CODE BEGIN 3 */
```

```
}
```

```
/* USER CODE END 3 */
```

```
}
```

```
/**
```

```
 * @brief System Clock Configuration
```

```
 * @retval None
```

```
 */
```

```
void SystemClock_Config(void)
```

```
{
```

```
    RCC_OscInitTypeDef RCC_OscInitStruct = {0};
```

```
    RCC_ClkInitTypeDef RCC_ClkInitStruct = {0};
```

```
/** Configure the main internal regulator output voltage
```

```
 */
```

```
    HAL_RCC_PWR_CLK_ENABLE();
```

```
    HAL_PWR_VOLTAGESCALING_CONFIG(PWR_REGULATOR_VOLTAGE_SCALE3);
```

```
/** Initializes the RCC Oscillators according to the specified parameters
```

```
 * in the RCC_OscInitTypeDef structure.
```

```
 */
```

```

RCC_OscInitStruct.OscillatorType = RCC_OSCILLATORTYPE_HSI;
RCC_OscInitStruct.HSISState = RCC_HSI_ON;
RCC_OscInitStruct.HSICalibrationValue = RCC_HSICALIBRATION_DEFAULT;
RCC_OscInitStruct.PLL.PLLState = RCC_PLL_NONE;

if (HAL_RCC_OscConfig(&RCC_OscInitStruct) != HAL_OK)
{
    Error_Handler();
}

/** Initializes the CPU, AHB and APB buses clocks
*/

RCC_ClkInitStruct.ClockType = RCC_CLOCKTYPE_HCLK|RCC_CLOCKTYPE_SYSCLK
                               |RCC_CLOCKTYPE_PCLK1|RCC_CLOCKTYPE_PCLK2;

RCC_ClkInitStruct.SYSCLKSource = RCC_SYSCLKSOURCE_HSI;
RCC_ClkInitStruct.AHBCLKDivider = RCC_SYSCLK_DIV1;
RCC_ClkInitStruct.APB1CLKDivider = RCC_HCLK_DIV1;
RCC_ClkInitStruct.APB2CLKDivider = RCC_HCLK_DIV1;

if (HAL_RCC_ClockConfig(&RCC_ClkInitStruct, FLASH_LATENCY_0) != HAL_OK)
{
    Error_Handler();
}

}

/**
 * @brief TIM2 Initialization Function
 * @param None
 * @retval None
 */

static void MX_TIM2_Init(void)
{

    /* USER CODE BEGIN TIM2_Init 0 */

    /* USER CODE END TIM2_Init 0 */

    TIM_ClockConfigTypeDef sClockSourceConfig = {0};
    TIM_MasterConfigTypeDef sMasterConfig = {0};

```

```

TIM_OC_InitTypeDef sConfigOC = {0};

/* USER CODE BEGIN TIM2_Init 1 */

/* USER CODE END TIM2_Init 1 */

htim2.Instance = TIM2;
htim2.Init.Prescaler = 0;
htim2.Init.CounterMode = TIM_COUNTERMODE_UP;
htim2.Init.Period = TIM2_Ticks - 1;
htim2.Init.ClockDivision = TIM_CLOCKDIVISION_DIV1;
htim2.Init.AutoReloadPreload = TIM_AUTORELOAD_PRELOAD_DISABLE;

if (HAL_TIM_Base_Init(&htim2) != HAL_OK)
{
    Error_Handler();
}

sClockSourceConfig.ClockSource = TIM_CLOCKSOURCE_INTERNAL;

if (HAL_TIM_ConfigClockSource(&htim2, &sClockSourceConfig) != HAL_OK)
{
    Error_Handler();
}

if (HAL_TIM_OC_Init(&htim2) != HAL_OK)
{
    Error_Handler();
}

sMasterConfig.MasterOutputTrigger = TIM_TRGO_RESET;
sMasterConfig.MasterSlaveMode = TIM_MASTERSLAVEMODE_DISABLE;

if (HAL_TIMEx_MasterConfigSynchronization(&htim2, &sMasterConfig) != HAL_OK)
{
    Error_Handler();
}

sConfigOC.OCMode = TIM_OCMODE_TIMING;
sConfigOC.Pulse = 0;
sConfigOC.OCpolarity = TIM_OCPOLARITY_HIGH;
sConfigOC.OCFastMode = TIM_OCFAST_DISABLE;

if (HAL_TIM_OC_ConfigChannel(&htim2, &sConfigOC, TIM_CHANNEL_1) != HAL_OK)
{
    Error_Handler();
}

```

```

/* USER CODE BEGIN TIM2_Init 2 */

/* TIM2_CH1 DMA Init */

HAL_RCC_DMA1_CLK_ENABLE();

hdma_tim2_ch1.Instance = DMA1_Stream5;
hdma_tim2_ch1.Init.Channel = DMA_CHANNEL_3;          // TIM2_CH1 is on channel 3
hdma_tim2_ch1.Init.Direction = DMA_MEMORY_TO_PERIPH; // Memory -> TIM3->CCR3
hdma_tim2_ch1.Init.PeriphInc = DMA_PINC_DISABLE;    // Peripheral address fixed
hdma_tim2_ch1.Init.MemInc = DMA_MINC_ENABLE;        // Memory address increments
hdma_tim2_ch1.Init.PeriphDataAlignment = DMA_PDATAALIGN_WORD;
hdma_tim2_ch1.Init.MemDataAlignment = DMA_MDATAALIGN_WORD;
hdma_tim2_ch1.Init.Mode = DMA_CIRCULAR;             // Repeat LUT automatically
hdma_tim2_ch1.Init.Priority = DMA_PRIORITY_HIGH;
hdma_tim2_ch1.Init.FIFOMode = DMA_FIFOMODE_DISABLE;

if (HAL_DMA_Init(&hdma_tim2_ch1) != HAL_OK)
{
    Error_Handler();
}

/* Link DMA handle to TIM2 handle */
__HAL_LINKDMA(&htim2, hdma[TIM_DMA_ID_CC1], hdma_tim2_ch1);

/* USER CODE END TIM2_Init 2 */

}

/**
 * @brief TIM3 Initialization Function
 * @param None
 * @retval None
 */
static void MX_TIM3_Init(void)
{

/* USER CODE BEGIN TIM3_Init 0 */

/* USER CODE END TIM3_Init 0 */

```



```

TIM_ClockConfigTypeDef sClockSourceConfig = {0};

TIM_MasterConfigTypeDef sMasterConfig = {0};

TIM_OC_InitTypeDef sConfigOC = {0};

/* USER CODE BEGIN TIM3_Init 1 */

/* USER CODE END TIM3_Init 1 */

htim3.Instance = TIM3;

htim3.Init.Prescaler = 0;

htim3.Init.CounterMode = TIM_COUNTERMODE_UP;

htim3.Init.Period = 4095;

htim3.Init.ClockDivision = TIM_CLOCKDIVISION_DIV1;

htim3.Init.AutoReloadPreload = TIM_AUTORELOAD_PRELOAD_DISABLE;

if (HAL_TIM_Base_Init(&htim3) != HAL_OK)

{

    Error_Handler();

}

sClockSourceConfig.ClockSource = TIM_CLOCKSOURCE_INTERNAL;

if (HAL_TIM_ConfigClockSource(&htim3, &sClockSourceConfig) != HAL_OK)

{

    Error_Handler();

}

if (HAL_TIM_PWM_Init(&htim3) != HAL_OK)

{

    Error_Handler();

}

sMasterConfig.MasterOutputTrigger = TIM_TRGO_RESET;

sMasterConfig.MasterSlaveMode = TIM_MASTERSLAVEMODE_DISABLE;

if (HAL_TIMEx_MasterConfigSynchronization(&htim3, &sMasterConfig) != HAL_OK)

{

    Error_Handler();

}

sConfigOC.OCMode = TIM_OCMODE_PWM1;

sConfigOC.Pulse = 0;

sConfigOC.OCpolarity = TIM_OCPOLARITY_HIGH;

sConfigOC.OCFastMode = TIM_OCFAST_DISABLE;

if (HAL_TIM_PWM_ConfigChannel(&htim3, &sConfigOC, TIM_CHANNEL_3) != HAL_OK)

{

```

```

    Error_Handler();
}

/* USER CODE BEGIN TIM3_Init 2 */

/* USER CODE END TIM3_Init 2 */
HAL_TIM_MspPostInit(&htim3);

}

/**
 * Enable DMA controller clock
 */

static void MX_DMA_Init(void)
{

    /* DMA controller clock enable */
    HAL_RCC_DMA1_CLK_ENABLE();

    /* DMA interrupt init */
    /* DMA1_Stream5_IRQn interrupt configuration */
    HAL_NVIC_SetPriority(DMA1_Stream5_IRQn, 0, 0);
    HAL_NVIC_EnableIRQ(DMA1_Stream5_IRQn);

}

/**
 * @brief GPIO Initialization Function
 * @param None
 * @retval None
 */

static void MX_GPIO_Init(void)
{
    GPIO_InitTypeDef GPIO_InitStruct = {0};

    /* USER CODE BEGIN MX_GPIO_Init_1 */

    /* USER CODE END MX_GPIO_Init_1 */

    /* GPIO Ports Clock Enable */

```

```

    HAL_RCC_GPIOH_CLK_ENABLE();

    HAL_RCC_GPIOA_CLK_ENABLE();

    HAL_RCC_GPIOB_CLK_ENABLE();

// -----
// LCD pins configuration
// -----
// Configure PC14 (RS) and PC15 (E) as output push-pull
GPIO_InitStruct.Pin = GPIO_PIN_14 | GPIO_PIN_15;
GPIO_InitStruct.Mode = GPIO_MODE_OUTPUT_PP;
GPIO_InitStruct.Pull = GPIO_NOPULL;
GPIO_InitStruct.Speed = GPIO_SPEED_FREQ_LOW;
HAL_GPIO_Init(GPIOC, &GPIO_InitStruct);

// Configure PB8 (D4) and PB9 (D5) as output push-pull
GPIO_InitStruct.Pin = GPIO_PIN_8 | GPIO_PIN_9;
HAL_GPIO_Init(GPIOB, &GPIO_InitStruct);

// Configure PA12 (D6) and PA15 (D7) as output push-pull
GPIO_InitStruct.Pin = GPIO_PIN_12 | GPIO_PIN_15;
HAL_GPIO_Init(GPIOA, &GPIO_InitStruct);

// Set all LCD pins LOW initially
HAL_GPIO_WritePin(GPIOC, GPIO_PIN_14 | GPIO_PIN_15, GPIO_PIN_RESET);
HAL_GPIO_WritePin(GPIOB, GPIO_PIN_8 | GPIO_PIN_9, GPIO_PIN_RESET);
HAL_GPIO_WritePin(GPIOA, GPIO_PIN_12 | GPIO_PIN_15, GPIO_PIN_RESET);

// -----
// Button0 configuration (PA0)
// -----
GPIO_InitStruct.Pin = Button0_Pin;
GPIO_InitStruct.Mode = GPIO_MODE_IT_RISING; // Interrupt on rising edge
GPIO_InitStruct.Pull = GPIO_PULLUP; // Use pull-up resistor
HAL_GPIO_Init(GPIOA, &GPIO_InitStruct);

// Enable and set EXTI line 0 interrupt priority
HAL_NVIC_SetPriority(EXTI0_IRQn, 2, 0);

```

```

    HAL_NVIC_EnableIRQ(EXTI0_IRQn);

    /* USER CODE END MX_GPIO_Init_2 */
}

/* USER CODE BEGIN 4 */
void EXTI0_IRQHandler(void) {

    // TODO: Debounce using HAL_GetTick()
    current_tick = HAL_GetTick();

    if (current_tick - last_button_press < 1000){
        HAL_GPIO_EXTI_IRQHandler(Button0_Pin); //clear interrupt flags
        return;
    }

    last_button_press = current_tick;

    // TODO: Disable DMA transfer and abort IT, then start DMA in IT mode with new LUT and
    re-enable transfer

    // HINT: Consider using C's "switch" function to handle LUT changes
    __HAL_TIM_DISABLE_DMA(&htim2, TIM_DMA_CC1);
    HAL_DMA_Abort_IT(&hdma_tim2_ch1);

    current_waveform = (current_waveform + 1) % 6;

    switch(current_waveform){
        case 0:
            selected_lut = Sin_LUT;
            waveform_name = "Sine";
            frequency = F_SIGNAL;
            ns_to_use = NS_WAVE;
            break;

        case 1:
            selected_lut = Saw_LUT;
            waveform_name = "Sawtooth";
            frequency = F_SIGNAL;

```

```

        ns_to_use = NS;

        break;

    case 2:

        selected_lut = Triangle_LUT;

        waveform_name = "Triangle";

        frequency = F_SIGNAL;

        ns_to_use = NS;

        break;

    case 3:

        selected_lut = Piano_LUT;

        waveform_name = "Piano";

        frequency = F_SIGNAL;

        ns_to_use = NS_WAVE;

        break;

    case 4:

        selected_lut = Guitar_LUT;

        waveform_name = "Guitar";

        frequency = F_SIGNAL/10;

        ns_to_use = NS_WAVE;

        break;

    case 5:

        selected_lut = Drum_LUT;

        waveform_name = "Drums";

        frequency = F_SIGNAL/10;

        ns_to_use = NS_WAVE;

        break;

    default:

        selected_lut = Sin_LUT;

        waveform_name = "Sine";

        frequency = F_SIGNAL/10;

        ns_to_use = NS;

}

```

```

TIM2_Ticks = TIM2CLK / (ns_to_use * frequency);

htim2.Init.Period = TIM2_Ticks - 1;

HAL_TIM_Base_Init(&htim2);


lcd_command(CLEAR);

lcd_putstr(waveform_name);


HAL_DMA_Start_IT(&hdma_tim2_ch1, (uint32_t)selected_lut, DestAddress, NS);


__HAL_TIM_ENABLE_DMA(&htim2, TIM_DMA_CC1);


HAL_GPIO_EXTI_IRQHandler(Button0_Pin);
}

/* USER CODE END 4 */

/**
 * @brief This function is executed in case of error occurrence.
 * @retval None
 */
void Error_Handler(void)
{
    /* USER CODE BEGIN Error_Handler_Debug */

    /* User can add his own implementation to report the HAL error return state */
    __disable_irq();

    while (1)
    {

    }

    /* USER CODE END Error_Handler_Debug */
}

#ifdef USE_FULL_ASSERT
/**
 * @brief Reports the name of the source file and the source line number
 * where the assert_param error has occurred.
 * @param file: pointer to the source file name
 * @param line: assert_param error line source number
 * @retval None
 */
void assert_failed(uint8_t *file, uint32_t line)

```

```
{  
    /* USER CODE BEGIN 6 */  
    /* User can add his own implementation to report the file name and line number,  
       ex: printf("Wrong parameters value: file %s on line %d\r\n", file, line) */  
    /* USER CODE END 6 */  
}  
#endif /* USE_FULL_ASSERT */
```